Predicting ICLR 2025 Peer-Review Decisions with Bayesian Models

Project Assignment 1: Exploring the Data and Thinking about Models

CSE 5825 Bayesian Machine Learning — Jiazheng Li

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This project asks whether the information in peer reviews can predict the decision on an ICLR 2025 submission (reject, or accept as a poster, spotlight or oral), and in particular whether the reviews carry information *beyond the mean rating* (Project 0). Each paper gets three to five reviews. Each reviewer picks an overall rating from only six options, $\{1, 3, 5, 6, 8, 10\}$, where 5 means "marginally below the acceptance threshold" and 6 means "marginally above". An area chair (AC) then reads the reviews and the discussion and recommends a decision. In this assignment I explore the data with this question in mind and look at three kinds of information beyond the mean: how the ratings fall around the acceptance threshold, how ratings change during the rebuttal, and how confident the reviewers are. All code is at `https://github.com/lijiazheng0917/bml-project`.

1 Working with data

1.1 The data set

What the data are. I combine three public releases built from OpenReview (Table 1). All three cover the same 11,672 ICLR 2025 submissions and are joined by the OpenReview paper ID.

Source	What I use	How it was collected
ICLR Insights (Kargaran et al., 2025)	For every reviewer: overall rating, confidence (1–5), and sub-scores for soundness, presentation and contribution (1–4), each *before and after* the rebuttal	Daily snapshots of OpenReview during the review period
ReviewerToo release	Final decision, primary area, title, abstract, review text, meta-review	Scraped from OpenReview after the decisions
Paper Copilot list	Submission status, including withdrawals	Scraped from OpenReview

Table 1: Data sources (all CC BY 4.0).

How the data were collected. ICLR runs its review process on OpenReview, where reviews, ratings and decisions are public. OpenReview only shows the latest version of a review, so a rating is lost once the reviewer updates it. The ICLR Insights team saved a snapshot every day during the review period and matched each reviewer across snapshots. This is why we can see each reviewer's rating before and after the rebuttal. They also give a "tracing score" that says how sure the match is.

Building the analysis sample. Table 2 shows the steps. ReviewerToo labels withdrawn papers as "reject", so I use Paper Copilot to separate them. A paper withdrawn *before* any AC decision (no meta-review) is removed. I keep 109 papers that the AC rejected and the authors withdrew afterwards, and 5 papers withdrawn after acceptance, because the AC did decide them. Finally, I randomly split the papers into a training set (80%) and a test set (20%), keeping the mix of decisions the same. **All figures in this report use only the training set.** The test set stays untouched so that the hypotheses in Section 2.3 can be tested fairly later.

Step	Papers
All submissions	11,672
removed: withdrawn before a decision	−2,874
removed: desk rejected, or conflicting labels	−71
Decided by an AC (reject 5,019; poster 3,115; spotlight 380; oral 213)	8,727
with at least two usable reviewer records (*analysis sample*)	8,655
training set, 80% (accept rate 42.5%)	6,925
test set, 20%	1,730

Table 2: From all submissions to the analysis sample.

Why the data interest me. Peer review decides which work gets attention, and many careers depend on it. Yet it is not clear how several reviews turn into one decision, or whether the mean rating, the number that authors and reviewers usually look at, tells the whole story. ICLR is one of the few venues where all reviews, ratings and decisions are public, and the ICLR Insights snapshots even show how each rating changed, so this step can be studied directly. As someone who submits to these conferences, I want to know what actually drives the decision.

Possible biases.

1. **Withdrawn papers are missing.** About a quarter of all submissions were withdrawn before a decision, and not at random: their mean initial rating is 4.0, against 5.0 for decided papers, and more than half of the papers with an initial mean below 4 were withdrawn. The results describe papers whose authors chose to stay in the process.
2. **One conference, one year.** The rating scale, reviewers and AC guidelines are specific to ICLR 2025.
3. **Reviewers are not assigned at random.** Reviewers bid on papers and are matched by expertise, so ratings and confidence partly reflect who chose to review a paper.
4. **The AC sees more than we do.** ACs read the paper and the full discussion. If, for example, a low rating points to a real flaw that the AC also sees, this information is missing from our data and can explain part of any pattern we find.
5. **Rating histories are reconstructed.** For 1,379 papers in the analysis sample (16%), the tracing score is 2 or more, so the before/after match may be wrong.
6. **Discussion text is incomplete.** The text release keeps only top-level posts; the authors' replies under each review are missing. This assignment uses only ratings, but it limits later text analysis.
7. **Authors may not be anonymous.** Many papers are on arXiv, so some reviewers may know who wrote them. We cannot observe this.

1.2 Three visualizations

Each figure looks at one kind of information beyond the mean rating. The figures describe patterns in the training set; they are not tests.

Figure 1: same mean, different ratings. *What it shows.* If ACs used only the mean rating, all dots in a panel would sit at the same height. They do not. Within each panel, the dots are ordered by colour, not by spread: the more reviewers give 6 or higher, the higher the acceptance rate. At a mean of 5.25, papers rated 3,6,6,6 were accepted 35% of the time, but papers rated 5,5,5,6 only 3%. At a mean of 6.0, papers rated 6,6,6,6 were accepted 91% of the time, but 5,5,6,8 only 44%. *Why it is interesting.* A single low rating does not sink a paper: 3,6,6,6 does far better than 5,5,5,6, although it contains the lowest rating. This goes against my starting idea that one strong negative review works as a veto. It also refines the disagreement hypothesis of Project 0: at the same mean, the spread itself is not what matters; what matters is how many ratings fall on each side of the threshold. Decisions seem to follow the count of accept-side recommendations rather than the average of the numbers. The only hint in the other direction is at a mean of 6.25, where 3,6,8,8 (71%) is below 5,6,6,8 (84%), but the two intervals overlap. *What it tells us about the data.* In practice the rating scale does not behave like evenly spaced numbers: the step from 5 to 6 seems to matter much more than its size of one point.

Figure 2: what the rebuttal changes. *What it shows.* Rebuttals move ratings almost only upward: 29.7% of reviewers raised their rating, 1.5% lowered it, and 68.8% kept it (panel B). The most common change is 5→6 (3,904 reviewers), which is exactly the move from "below the threshold" to "above the threshold". Next come 6→8 (1,732) and 3→5 (1,422). Reviewers who started at 5 changed most often (42% raised). In panel A, the two curves lie on top of each other: at the same final mean, papers that got there through a rebuttal increase are accepted about as often as papers that had that mean from the start. *Why it is interesting.* This is the rebuttal-trajectory question from Project 0. Panel A suggests that the decision mostly follows where the ratings ended, not how they got there. But Figure 1 shows that the *number* of accept-side ratings matters, and panel B shows that rebuttals mostly create new accept-side ratings (5→6). The figure cannot tell whether such a converted rating counts as much as one that was on the accept side from the start. To see that, we must also hold the number of accept-side ratings fixed, which needs the model in Section 2.

Figure 3: how confidence is used. *What it shows.* Panel A: a reviewer's confidence says little about how close the reviewer is to the other reviewers. The distance is smallest at confidence 3 (1.25 points) and largest at confidence 5 (1.45 points). If high confidence meant high accuracy, the line would go down; instead it is U-shaped. Panel B: when the confident reviewers are more negative than the others, papers are accepted about 3–4 percentage points less often than their mean predicts; when the confident reviewers are more positive (0.1 to 0.3), about 3 points more often. The 30 papers above 0.3 give no clear signal. *Why it is interesting.* Together, the panels suggest that decisions lean toward the opinion of confident reviewers, even though confidence does not mark the reviewers who agree with their peers. This would be a mismatch between how confidence is used and how informative it is. The effect is small and most papers sit in the middle bin, so it needs careful testing.

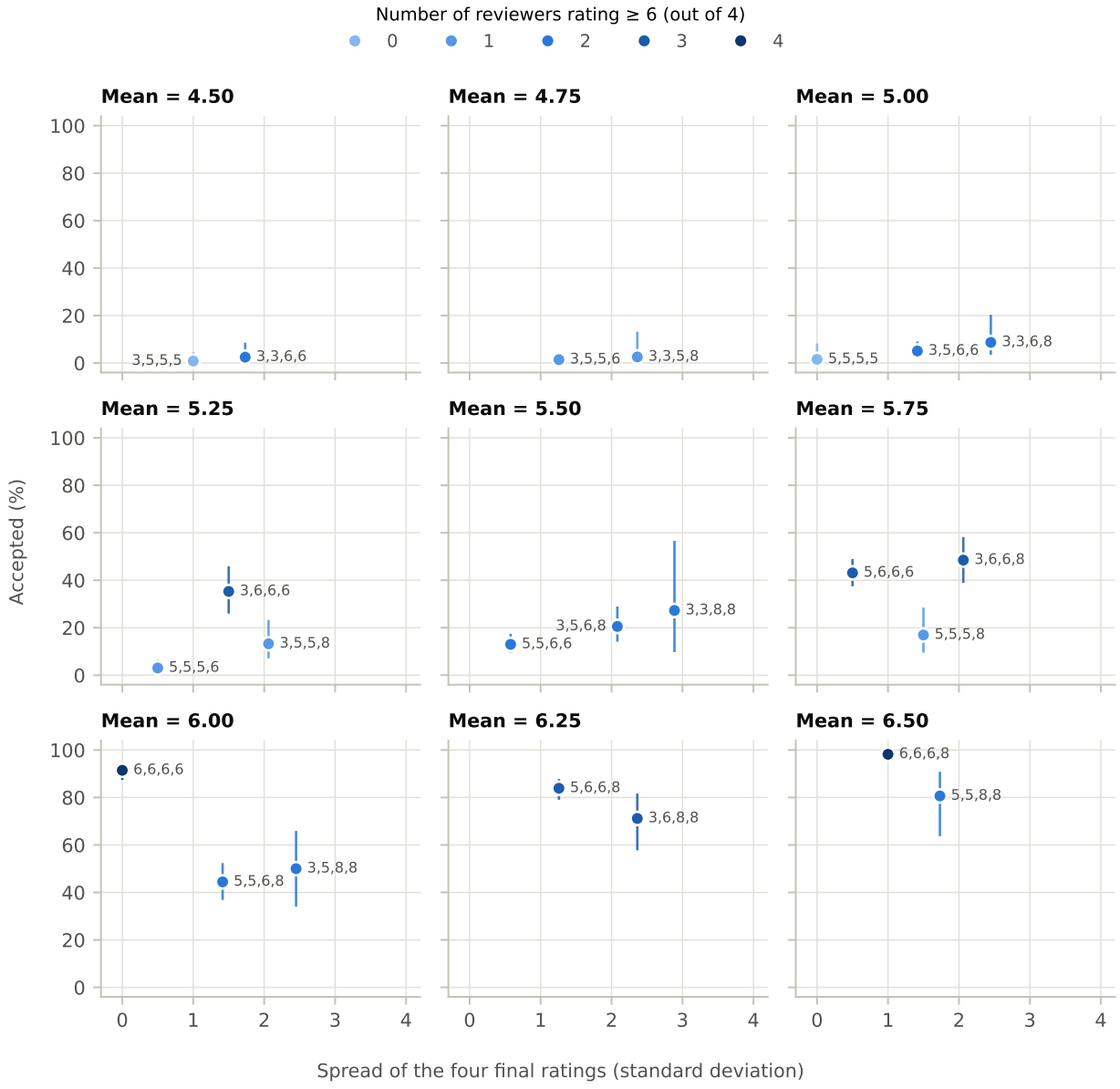


Figure 1: Acceptance rate for each combination of four final ratings, with one panel per mean rating (training set, papers with exactly four reviewers, $n = 4{,}656$). Each dot is one combination seen in at least 10 papers; the vertical line is a 95% confidence interval. The x-axis is the spread (standard deviation) of the four ratings. Darker dots have more reviewers on the accept side (rating ≥ 6).

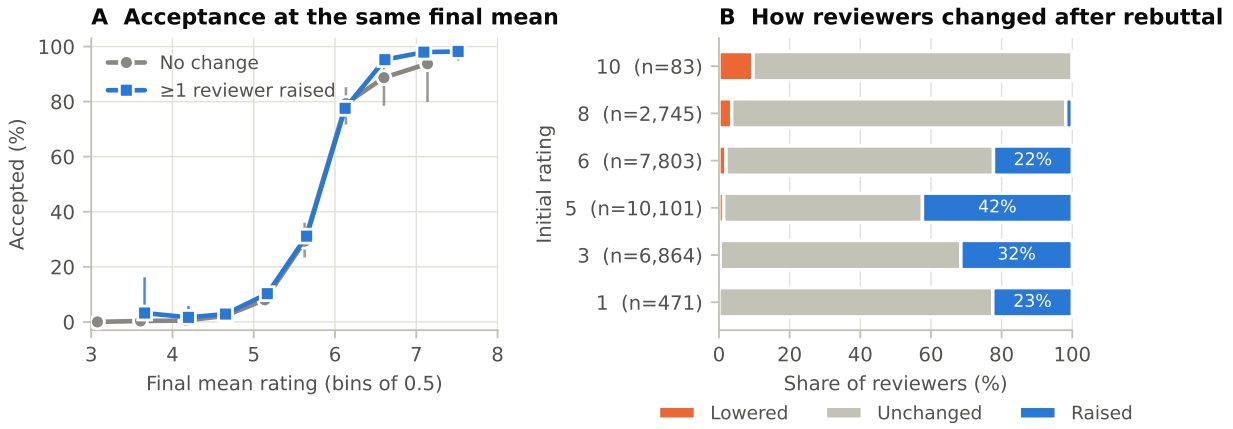


Figure 2: (A) Acceptance rate against the final mean rating, for papers where at least one reviewer raised the rating after the rebuttal (and nobody lowered it), and for papers where no rating changed. Final means are grouped in bins of 0.5; vertical lines are 95% confidence intervals. (B) How reviewers changed their rating after the rebuttal, by their initial rating. Training set.

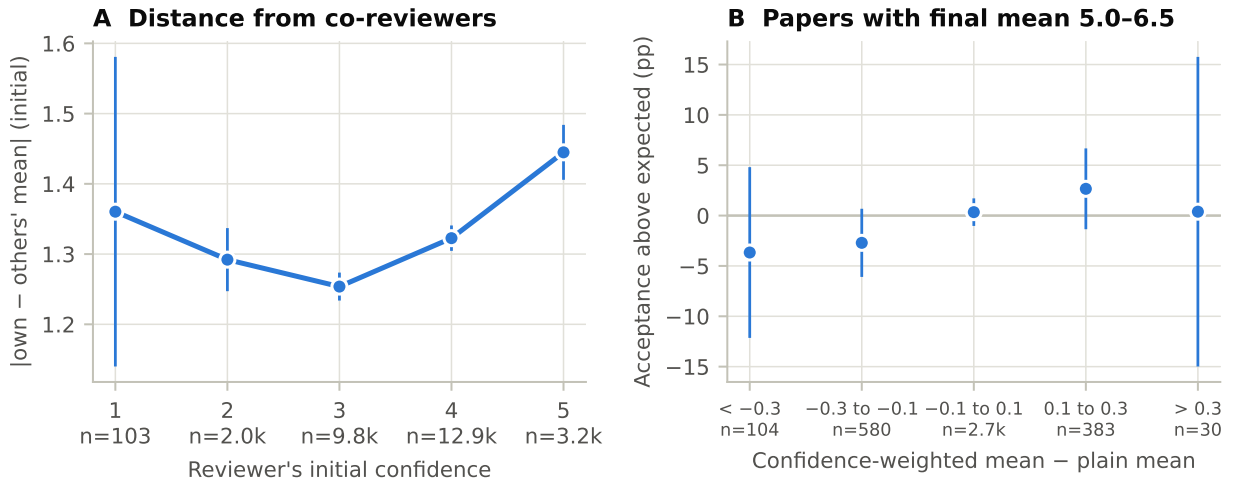


Figure 3: (A) Average distance between a reviewer's initial rating and the mean initial rating of the *other* reviewers of the same paper, by the reviewer's confidence. (B) Papers with a final mean between 5.0 and 6.5: actual acceptance rate minus the rate expected from the mean alone (the acceptance rate of all training papers with exactly the same mean), grouped by how much the confidence-weighted mean differs from the plain mean. Positive x-values mean that the more confident reviewers liked the paper more. Error bars are 95% confidence intervals.

2 Thinking about models

2.1 Observed variables and their relationships

Table 3 lists the observed variables for paper i and reviewer j. From them I compute the final mean rating $\bar{r}_i^{(1)}$ and the number of accept-side ratings $V_i = \#\{j : r_{ij}^{(1)} \geq 6\}$.

Symbol	Meaning	Values
$r_{ij}^{(0)}$, $r_{ij}^{(1)}$	overall rating before / after the rebuttal	$\{1, 3, 5, 6, 8, 10\}$
$c_{ij}^{(0)}$, $c_{ij}^{(1)}$	reviewer confidence before / after	1–5
$s_{ijk}^{(0)}$, $s_{ijk}^{(1)}$	sub-score k (soundness, presentation, contribution)	1–4
J_i	number of reviewers	mostly 3–5
a_i	primary research area	21 areas
Y_i	decision	reject, poster, spotlight, oral

Table 3: Observed variables.

Expected relationships.

- **Ratings and sub-scores move together.** The overall rating is most related to contribution (rank correlation 0.65), then soundness (0.59), then presentation (0.46).
- **Confidence is almost unrelated to the rating**; confident reviewers give slightly lower ratings (rank correlation -0.14).
- **Initial and final ratings are strongly related**: 69% of reviewers keep their rating.
- **The decision depends strongly on the final ratings** (Figure 1). The acceptance rate is a bit lower for papers with more reviewers (47% with three, 42% with four, 41% with five), possibly because extra reviewers are added to difficult papers.

Independence and conditional independence.

- **Different papers** are treated as independent given the shared parameters. This is not exactly true (a reviewer or an AC handles several papers), but reviewer and AC IDs are not linked across papers, so we cannot model it.
- **Initial ratings of the same paper** are correlated, because they respond to the same paper. But reviewers write their first reviews without seeing each other's, so I expect them to be *independent given the paper's hidden quality* q_i: $r_{ij}^{(0)} \perp r_{ij'}^{(0)} \mid q_i$. The correlation is weak: only 15% of the variation in initial ratings is between papers (intraclass correlation, ICC $= 0.15$); the rest is disagreement among reviewers of the same paper.
- **Final ratings are not independent given quality**, because reviewers read each other's reviews and the discussion before updating. The data agree: the ICC rises from 0.15 to 0.33 after the rebuttal, so reviewers move toward each other.
- **The decision.** Each hypothesis in Section 2.3 claims that one conditional independence *fails*: H1, $Y_i \perp V_i \mid \bar{r}_i^{(1)}$ (votes matter beyond the mean); H2, $Y_i \perp r_{i\cdot}^{(0)} \mid r_{i\cdot}^{(1)}$ (initial ratings matter beyond the final ones); H3, $Y_i \perp c_{i\cdot} \mid r_{i\cdot}^{(1)}$ (confidence matters beyond the ratings).

2.2 Hidden variables

Several important quantities are not observed. Figure 4 shows how they connect to the data.

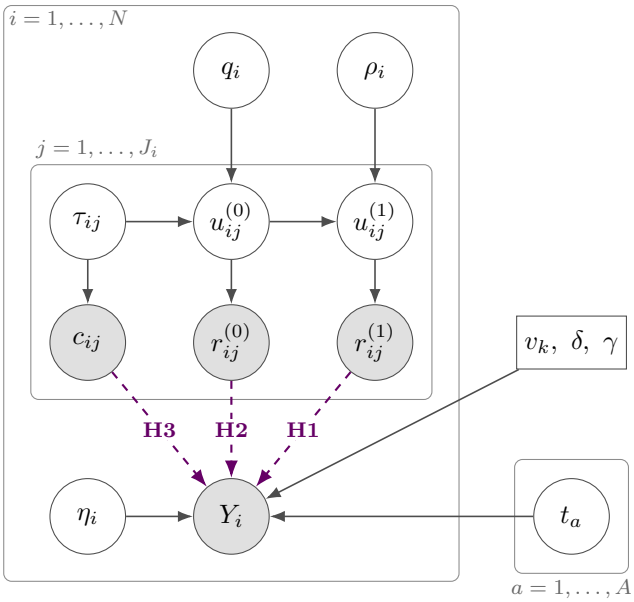


Figure 4: Graphical model. Grey circles are observed, white circles are hidden, and the square box holds the shared parameters of the decision model. Rounded rectangles are plates (repeated for each reviewer, paper and area). Dashed purple arrows are the links that the three hypotheses are about.

- q_i — **paper quality**: what all reviewers try to judge. It explains why ratings of the same paper are correlated, and why initial ratings are independent once we know it.
- $u_{ij}^{(0)}, u_{ij}^{(1)}$ — **a reviewer's opinion on a continuous scale**, before and after the rebuttal. The rating is this opinion cut into six bins, and the cut points are learned, so the model does not assume that the step from 5 to 6 is worth exactly one point. It also explains why an opinion can change while the rating stays the same.
- τ_{ij} — **reviewer precision**: how close the reviewer's opinion is to the paper's quality. The reported confidence is the reviewer's own guess of τ_{ij}, and the model lets us check how good that guess is.
- ρ_i — **rebuttal effect**: how much the authors' response moves all reviewers of the paper. Reviewers moving toward each other is a second, within-paper effect that is not drawn in Figure 4.
- t_a — **acceptance threshold of research area** a, with a shared prior so that small areas borrow information from large ones.
- η_i — **AC private information**: what the AC learns from the paper and the discussion beyond the ratings. We cannot measure it, so the model treats it as random noise with a fixed scale, and all other effects are measured relative to it.

The decision model. This is the likelihood $p(Y_i \mid \mathbf{x}_i, \theta)$ from Project 0, now built from the individual ratings instead of their mean and standard deviation:

$$\Pr(Y_i = \text{accept}) = \text{logistic}\big(S_i - t_{a(i)}\big), \qquad S_i = \sum_j w_{ij}\Big[(1-\delta)\, v\big(r_{ij}^{(1)}\big) + \delta\, v\big(r_{ij}^{(0)}\big)\Big], \qquad w_{ij} \propto e^{\gamma\, c_{ij}^{(1)}},$$

where the weights w_{ij} of a paper sum to one and the logistic noise is the AC's private information η_i. Three shared parameters describe how the ratings are combined: $v(k)$ is the value given to a rating of k (one free number per rating level); δ is the weight on the *initial* ratings; γ says how much more confident reviewers count. If $v(k)$ is a straight line in k, $\delta = 0$ and $\gamma = 0$, the model is

simply "use the mean final rating", which is the baseline of Project 0. Spotlight and oral are added as higher thresholds on the same score, which gives the four decision classes of Project 0.

To ask whether this way of combining ratings is *reasonable*, I also fit a measurement model to the initial ratings: $u_{ij}^{(0)} = q_i + \varepsilon_{ij}$ with $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2(c_{ij}))$, and $r_{ij}^{(0)}$ is $u_{ij}^{(0)}$ cut into bins. Because initial ratings are independent given q_i, their agreement tells us how much each rating level and each confidence level says about quality. From this we can work out how an *ideal AC* — one that ranks papers by their expected quality given all ratings — would weigh the ratings, and compare it with the fitted v, δ and γ.

2.3 Hypotheses

The three hypotheses update those of Project 0 using what the figures show: H1 refines the disagreement hypothesis, H2 refines the rebuttal-trajectory hypothesis, and H3 replaces the discussion hypothesis, which cannot be tested because the reply counts in the data are all zero (Project 0). Each hypothesis has two parts: (a) a **pattern** that should also appear in the held-out test set, and (b) a **deeper question**: are the ratings combined the way the ideal AC above would combine them? Part (b) cannot be read off a figure; it needs the hidden-variable model.

H1: accept-side ratings matter beyond the mean. (a) At the same final mean rating, the chance of acceptance rises with the number of reviewers who rate the paper 6 or higher; a single low rating (≤ 3) does not work as a veto. (b) In the decision model, the gap $v(6) - v(5)$ is larger than the gaps $v(5) - v(3)$ and $v(8) - v(6)$, and larger than an ideal AC would use. In plain words: a rating works mainly as an accept/reject vote. *Why I expect it:* Figure 1, and the rating labels themselves ("marginally below/above the threshold") read like votes. *What would count against it:* adding the vote count, or free values $v(k)$, does not improve predictions on the test set; or the fitted $v(k)$ lie close to a straight line.

H2: rebuttal trajectory — converted votes count less. (a) Among papers with the same final mean and the same number of accept-side ratings, papers in which a reviewer moved from below 6 to 6 or higher during the rebuttal are accepted less often than papers whose accept-side ratings were there from the start. (b) $\delta > 0$: the decision keeps some weight on the initial ratings. An ideal AC should do this, because initial ratings are independent opinions, while final ratings partly reflect reviewers moving toward each other (ICC 0.15 $\rightarrow$ 0.33). *Why I expect it:* 5$\rightarrow$6 is the most common change (Figure 2B), and ACs can see the rating history and the discussion. *What would count against it:* δ close to 0 (ACs use only the final ratings, as Figure 2A suggests at the level of the mean), or $\delta < 0$ (ACs reward a successful rebuttal).

H3: reviewer confidence gets too much weight. (a) At the same final mean and number of accept-side ratings, papers whose confident reviewers are more positive than their less confident reviewers are accepted more often ($\gamma > 0$). (b) In the measurement model, a reviewer's noise $\sigma(c)$ hardly shrinks with confidence, so the weight on confidence is larger than an ideal AC would use. *Why I expect it:* Figure 3B for part (a) and Figure 3A for part (b). *What would count against it:* γ close to 0; or $\sigma(c)$ clearly shrinks with confidence, in which case listening to confident reviewers is reasonable.

How I will test them later.
1. The hypotheses and the analysis plan are fixed now, before anyone looks at the test set.

2. Fit the decision model and the measurement model with MCMC on the training set, and report posterior intervals for $v(k)$, δ, γ and $\sigma(c)$.
3. Compare versions of the decision model (mean only, as in the Project 0 baseline; mean plus vote count; free $v(k)$; with and without δ and γ) by their log-likelihood on the 20% test set.
4. Posterior predictive check: simulate decisions from the fitted model and check that they reproduce Figure 1.
5. Repeat the analysis on ICLR 2024, which ICLR Insights also covers with the same rating scale.
6. Check sensitivity: use only reviewers with a tracing score of 1 or less, and treat the rejected-then-withdrawn papers differently.